

Yulong Zhao

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Education

University of Arizona — PhD in Statistics	2023 – Expected 05/2028
University of Wisconsin–Madison — MS in Statistics and Data Science	2022 – 2023
University of Science and Technology of China — BS in Mathematics and Applied Mathematics	2018 – 2022

Research Experience

Real-Time Early Warning with a Controlled False-Alarm Rate 2026 – Present

University of Arizona | Advisor: Prof. Xueying Tang

- Built a **real-time risk-scoring** system that reads each user's partial action sequence in OECD PISA problem-solving tasks and flags users heading toward failure while they are still working.
- Designed a **false-alarm-controlled** alert rule: thresholds are calibrated on the deployed trigger (2 hits in 4 steps), capping false alarms on users who would succeed at $\leq 5\%$.
- Benchmarked ~ 370 GRU, temporal CNN, and Transformer configurations against leakage-free logistic baselines; the best temporal CNN caught **98.9%** of failing users at precision ≥ 0.98 , with false alarms $\leq 4\%$ on all 3 tasks (GRU: 85.6%).

Time-Aware Modeling of User Action Logs Sept 2023 – Present

University of Arizona | Advisor: Prof. Xueying Tang

- Developed an **interpretable time-aware model** for process data (timestamped action logs) that tests, for each action, whether its effect on the outcome is absent, constant, or depends on *when* it happens (functional regression, adaptive group LASSO).
- On 19,089 events from 2,668 test-takers in OECD PISA 2012, raised cross-validated accuracy from 93.5% to **96.1%** and **halved log loss** ($0.237 \rightarrow 0.118$) vs. a count-feature baseline, winning on all 10 folds.
- Showed the gain comes from actions whose effect **flips sign with timing** (early exploration vs. final choice), which count features cannot capture; quantified uncertainty with 300 bootstrap refits and 1,200 simulations.

Latent Skill and Hidden-State Modeling of User Behavior 2026 – Present

University of Arizona | Advisor: Prof. Xueying Tang

- Designed a **structured latent-variable framework** that separates stable user skills from evolving problem-solving states, generating each step as hidden state $\rightarrow$ interface choice $\rightarrow$ action (a skill-driven **hidden Markov model**).
- Made every component interpretable: sparse skill-to-behavior maps for state transitions, interface choice, and action choice show which skill changes which behavior, learned with sparse-group LASSO and refit by maximum likelihood.
- Implemented estimation from scratch in Python (forward-backward EM, multi-start) and searched 45 candidate dependency structures by BIC across 580+ HPC jobs, on 1,340 users of an email-and-spreadsheet task.

Time-Series Forecasting and Regime-Shift Detection May 2024 – Present

University of Arizona | IMSI Interdisciplinary Research Cluster

- Built a **walk-forward forecasting** pipeline (RBF-SVR, time-ordered CV, no future-information leakage) on PCA-compressed features from 8 environmental series; beat the historical-mean baseline in **18/18** settings with 38% lower MSE.
- Developed a **regime-shift detector** across 28 series pairs, tested against 4,999 autocorrelation-preserving null simulations with Holm correction; simulation confirmed a false-positive rate $\leq 2/50$.
- Stress-tested findings under 49 data perturbations, which flagged 1 of 6 candidate shifts as fragile; delivered visual reports to geoscience collaborators.

Publications and Presentations

Zhao, Y., & Tang, X. Scalar-on-process model for educational assessment: modeling time-varying action effects with group LASSO regularization. *Manuscript ready for submission.*

Refereed oral presentation of the above, Symposium on Data Science & Statistics (SDSS), Milwaukee, WI, April 2026.

Skills

Programming: Python (NumPy, pandas, scikit-learn, Numba), R (Keras/TensorFlow), SQL, Git, Slurm/HPC, LaTeX.

Machine learning: sequence models (GRU, TCN, Transformer), regularized GLMs, kernel methods, latent-variable models/EM.

Statistics: threshold calibration, leakage-safe validation, bootstrap, multiple testing, simulation studies.

Awards

Research Fellowship (2024–26) & Incoming PhD Award, U. Arizona; Academic Excellence Award, UW–Madison; Outstanding Undergraduate Award, USTC.